

Daily Geometry Drill #1 — Answers & Investigations

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Section A Rapid Recognition

1. Find the gradient of the line $6x - 3y + 15 = 0$.

Answer: 2

Method: Rearrange to gradient–intercept form: $3y = 6x + 15 \Rightarrow y = 2x + 5$, so $m = 2$.

Investigate further: Implicit form $ax + by + c = 0$ has gradient $-a/b$; here $-\frac{6}{3} = 2$. **If the line is instead written $3y - 6x - 15 = 0$ (the same line), is the gradient still 2?** It must be — check the $-a/b$ rule still agrees.

2. Find the gradient of the line through $(1, 2)$ and $(5, 14)$.

Answer: 3

Method: $m = \frac{14-2}{5-1} = \frac{12}{4} = 3$.

Investigate further: Check with $(5 - 1, 14 - 2)$ swapped: $\frac{2-14}{1-5} = \frac{-12}{-4} = 3$ ✓ — the sign always cancels. **If the line through $(1, 2)$ and $(5, 14)$ is moved up by 3 (every y increased by 3), does its gradient change?** No — a vertical translation keeps the ratio of rises fixed.

3. A line of gradient 4 is perpendicular to a line of gradient m . Write down m .

Answer: $-\frac{1}{4}$

Method: Perpendicular gradients are negative reciprocals: $m = -\frac{1}{4}$, since $4 \cdot (-\frac{1}{4}) = -1$.

Investigate further: The product test $m_1 m_2 = -1$ is the fastest check. **Is the line $y = 4x$ parallel, perpendicular, or neither, to $y = -\frac{1}{4}x + 12$?** The constant term is a distraction — gradients alone decide.

4. Find the distance between $(2, 1)$ and $(5, 5)$.

Answer: 5

Method: $d = \sqrt{(5-2)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = 5$.

Investigate further: 3-4-5 right triangle in disguise. **The points $(-1, 0)$ and $(3, -3)$ are the same distance apart — can you see the Pythagorean triple before computing, by drawing the displacement vector?** The displacement $(4, -3)$ is a reordered 3-4-5.

5. Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.

Answer: $(7, 5)$

Method: Midpoint averages each coordinate: $\left(\frac{4+10}{2}, \frac{8+2}{2}\right) = (7, 5)$.

Investigate further: Check both halves are equal distances: $(7-4, 8-5) = (3, 3)$ and $(10-7, 5-2) = (3, 3) \checkmark$. **The midpoint of two points is the average of the two displacement vectors. What is the midpoint of $(4, 8)$ and $(-6, -2)$?** Negative coordinates behave the same way.

6. Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.

Answer: 2

Method: Use $\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$ at $(0, 0)$: $\frac{|-10|}{\sqrt{9+16}} = \frac{10}{5} = 2$.

Investigate further: The denominator 5 is the length of the normal vector $(3, 4)$. **The point of the line closest to the origin satisfies $(x, y) = t(3, 4)$ for some t — find it and confirm its distance is indeed 2.** Every point of the line's coordinate vector is a multiple of its normal.

7. A line has x -intercept 4 and y -intercept 6. Find its gradient.

Answer: $-\frac{3}{2}$

Method: The intercepts are the points $(4, 0)$ and $(0, 6)$: $m = \frac{6-0}{0-4} = -\frac{3}{2}$.

Investigate further: Reading right-to-left: this is the negative of $\frac{y\text{-intercept}}{x\text{-intercept}}$. **Write the line in intercept form $\frac{x}{4} + \frac{y}{6} = 1$ and rearrange to confirm $m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\frac{6}{4} = -\frac{3}{2}$ — the intercept form makes the gradient transparent.**

8. Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.

Answer: 6

Method: Right-angled at the origin: area $= \frac{1}{2} \cdot 4 \cdot 3 = 6$.

Investigate further: Equivalently the shoelace/midpoint-vector formula $\frac{1}{2}|(4)(3) - (0)(0)| = 6$. **Rotate this triangle by 90° about the origin: does the area change, and do you get the same shoelace number?** Reflections and rotations preserve area.

9. The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .

Answer: 5

Method: Rise per unit run is 2 (from $(1, 2)$ to $(2, 4)$); continuing to $y = 10$ climbs 6 more, so $k = 2 + \frac{6}{2} = 5$.

Investigate further: Formal check: $\frac{4-2}{2-1} = \frac{10-4}{k-2} \Rightarrow 2 = \frac{6}{k-2} \Rightarrow k = 5$. **What happens to k if the points are required to be collinear but the order is $(2, 4)$, $(1, 2)$, $(k, 10)$?** Collinearity is order-independent — all three gradients must agree.

10. Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Answer: 12

Method: Shoelace: $\frac{1}{2}|0 \cdot 4 + 3 \cdot 0 + 6 \cdot 0 - (0 \cdot 3 + 4 \cdot 6 + 0 \cdot 0)| = \frac{1}{2}|24| = 12$.

Investigate further: Bonus route: base = 6, height = 4, area = $\frac{1}{2} \cdot 6 \cdot 4 = 12$ ✓. **For a triangle with one vertex at the origin, shoelace collapses to $\frac{1}{2}|x_1y_2 - x_2y_1|$ with the other two points — prove that this equals $\frac{1}{2}|\det|$ of the two vectors.** That determinant is the signed area.

Section B Manipulation Drills

1. The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

A) $y = -\frac{3}{2}x + 1$

B) $y = \frac{3}{2}x - 2$

C) $y = \frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + 3$

Answer: A) $y = -\frac{3}{2}x + 1$

Method: Given line has gradient $\frac{2}{3}$ (rule $-a/b$), so perpendicular gradient is $-\frac{3}{2}$. Only option A has gradient $-\frac{3}{2}$.

Investigate further: Options B and C have the same gradient, and D the new “same slope” decoy — read *gradients only* through the $-\frac{3}{2}$ filter. **Which option is parallel to the given line, and which is merely reflection-symmetric in slope?** Perpendicularity and parallelism are decided by one number, the slope.

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

A) 1

B) 2

C) 3

D) 4

Answer: C) 3

Method: $\frac{|5(2)+12(3)-7|}{\sqrt{25+144}} = \frac{|10+36-7|}{13} = \frac{39}{13} = 3$.

Investigate further: The numerator is exactly how far the point sits from the line measured along a normal whose length is 13. **Two points are at perpendicular distance 3 from $5x + 12y - 7 = 0$ — what do they have in common?** They lie on $5x + 12y - 7 = \pm 39$, the two parallel satellite lines.

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

A) $(2, 5)$

B) $(3, 3)$

C) $(4, 4)$

D) $(5, 3)$

Answer: A) (2, 5)

Method: Midpoint (3, 4); segment gradient 1, so bisector gradient -1 : $y - 4 = -(x - 3)$, i.e. $x + y = 7$. Check options: $2 + 5 = 7$ ✓.

Investigate further: Only A satisfies $x + y = 7$. **A point is on the perpendicular bisector iff it is equidistant from the two endpoints — verify (2, 5) is distance $\sqrt{16}$ from both (1, 2) and (5, 6).** That's the defining property, worth internalising.

4. Find the area of the quadrilateral with vertices (0, 0), (4, 0), (5, 3), (1, 3), in that order.

- A) 10
B) 12
C) 14
D) 16

Answer: B) 12

Method: Shoelace in order: $\frac{1}{2}|0 \cdot 0 + 4 \cdot 3 + 5 \cdot 3 + 1 \cdot 0 - (0 \cdot 4 + 0 \cdot 5 + 3 \cdot 1 + 3 \cdot 0)| = \frac{1}{2}|0 + 12 + 15 + 0 - (0 + 0 + 3 + 0)| = \frac{1}{2}|24| = 12$.

Investigate further: The quadrilateral is a parallelogram (bases $\parallel$ via position vectors (4, 0) and (1, 3)), so area = $|\det| = |4(3) - 0(1)| = 12$ ✓. **Reorder the vertices as (0, 0), (5, 3), (4, 0), (1, 3): the same region, but the shoelace number warns you when points are listed in the wrong order.** Keep vertices in cyclic order.

5. The line through (1, 2) and (4, 5) crosses the y -axis at which value?

- A) 0
B) 1
C) 2
D) 3

Answer: B) 1

Method: Gradient $\frac{5-2}{4-1} = 1$; $y = x + c$ through (1, 2) gives $c = 1$.

Investigate further: Extension: at $x = 0$, $y = 1$. **Through the point-slope form, $y - 2 = 1(x - 1)$. If the line were instead the one through (1, 2) and (4, 8), the y -intercept changes — resolve c again.** Always plug the known point in, never guess c from the picture.

6. Find the perpendicular distance from (0, 1) to the line $6x - 8y + 2 = 0$.

Answer: $\frac{3}{5}$

Method: $\frac{|6(0) - 8(1) + 2|}{\sqrt{36 + 64}} = \frac{|-6|}{10} = \frac{3}{5}$.

Investigate further: Normal length $\sqrt{36 + 64} = 10$; numerator 6. **The line $6x - 8y + 2 = 0$ sits distance $\frac{1}{5}$ from the origin (check $\frac{|2|}{10}$). Since the y -axis point (0, 1) is 1 unit up from the origin, the net distance $|1 - \frac{1}{5}| = \frac{4}{5}$ differs from $\frac{3}{5}$ — why?** Because the normal (6, -8) is not vertical: only the projection along it counts.

7. Which of the following is NOT parallel to $y = 3x + 1$?

- A) $y = 3x + 5$
- B) $6x - 2y = 4$
- C) $3x - y = 2$
- D) $y = 2x + 1$

Answer: D) $y = 2x + 1$

Method: Parallel $\Leftrightarrow$ same gradient 3. A: $3\checkmark$; B: $6x - 2y = 4 \Rightarrow y = 3x - 2\checkmark$; C: $3x - y = 2 \Rightarrow y = 3x - 2\checkmark$; D: $y = 2x + 1$ has gradient 2.

Investigate further: Options B and C are disguises (divide by the y -coefficient first). **Show two distinct parallel lines never intersect, while any two non-parallel lines always do.** Parallelism is exactly “same slope, different intercept”.

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .

Answer: $(1, 2)$

Method: Through $(0, 3)$ with gradient -1 : $y = -x + 3$. Intersect: $x + 1 = -x + 3 \Rightarrow x = 1$, then $y = 2$.

Investigate further: The intersection of two perpendicular lines is the right-angle vertex of a right triangle. Verify $P = (1, 2)$ lies on both lines: $y = x + 1$ gives $2 = 1 + 1 \checkmark$ and $y = -x + 3$ gives $2 = -1 + 3 \checkmark$. **Through $(0, 3)$ draw the perpendicular and reflect the point where it meets $y = x + 1$ across that meeting point — what do you get, and how does it connect to the midpoint of $(0, 3)$ and $(2, 1)$?** Perpendicular lines meet exactly once; the reflect-and-extend trick answers the follow-up.

9. Which point is closest to the origin?

- A) $(1, 3)$
- B) $(2, 2)$
- C) $(4, 0)$
- D) $(0, 5)$

Answer: B) $(2, 2)$

Method: Distances: $\sqrt{(1, 3)} = \sqrt{10}$, $\sqrt{(2, 2)} = \sqrt{8}$, $\sqrt{(4, 0)} = 4$, $\sqrt{(0, 5)} = 5$; smallest is $\sqrt{8}$.

Investigate further: Comparing squared distances avoids surds: 10, 8, 16, 25. **Among points with x, y nonnegative integers and $x + y = 4$, which is closest to the origin?** Both squares and sum compare against $(2, 2)$.

10. Using the shoelace formula, find the area of the triangle with vertices $(1, 1)$, $(4, 2)$, $(2, 6)$.

Answer: 7

Method: $\frac{1}{2}|1 \cdot 2 + 4 \cdot 6 + 2 \cdot 1 - (1 \cdot 4 + 2 \cdot 2 + 6 \cdot 1)| = \frac{1}{2}|2 + 24 + 2 - (4 + 4 + 6)| = \frac{1}{2}|28 - 14| = 7$.

Investigate further: The 7 here is exactly the determinant area of the two vectors $(3, 1)$ and

$$(1, 5): \frac{1}{2}|3(5) - 1(1)| = \frac{1}{2}(14) = 7 \checkmark. \text{ In shoelace, matching – and cancelling – the repeated coordinate arithmetic is a good fastest-check habit. Confirm you did } \frac{1}{2}|(x_1y_2 + x_2y_3 + x_3y_1) - (y_1x_2 + y_2x_3 + y_3x_1)|.$$

Section C Substitution & Structure

(≤ 90 s each)

Competition-hard: hidden structure, non-routine use of the same tools. No question is a direct formula plug.

1. The lines $l_1 : y = 6 - 2x$ and $l_2 : y = \frac{1}{2}x + 3$ are perpendicular and intersect at P . The three lines l_1 , l_2 and the x -axis enclose a triangle T . What is the area of T ? (after TMUA 2017 Paper 1 Q3)

- A) $\frac{81}{5}$
 B) $\frac{54}{5}$
 C) $\frac{81}{10}$
 D) 27

Answer: A) $\frac{81}{5}$

Method: l_1 has x -intercept 3 ($6 - 2x = 0$), l_2 has x -intercept -6 ($x/2 + 3 = 0$). They meet where $6 - 2x = x/2 + 3 \Rightarrow 5x/2 = 3 \Rightarrow x = 6/5$, $y = 18/5$. Base on x -axis is $3 - (-6) = 9$, height is $y_P = 18/5$, so area $= \frac{1}{2} \cdot 9 \cdot \frac{18}{5} = \frac{81}{5}$. Shoelace on $(3, 0)$, $(-6, 0)$, $(6/5, 18/5)$ gives the same $81/5$.

Investigate further: The x -intercepts are read without solving the system; the height is the single y of the intersection. **Check with base-height on the x -axis vs. shoelace: both give $2 \cdot \text{area} = 81/5 \cdot 2$. If l_2 were instead $y = \frac{1}{2}x - 3$ (x -int 6), the base would be 3 and the area would collapse to $\frac{27}{10}$ — why does the sign of the intercept matter?**

2. The perpendicular bisector of the segment joining $A(2, -6)$ and $B(5, 4)$ meets the x -axis at P . What is the x -coordinate of P ? (after TMUA Specimen Paper 1 Q3)

- A) $\frac{1}{6}$
 B) $\frac{1}{3}$
 C) $\frac{19}{5}$
 D) $\frac{41}{6}$

Answer: A) $\frac{1}{6}$

Method: Midpoint $M = (7/2, -1)$, gradient $AB = (4 - (-6))/(5 - 2) = 10/3$, so perp gradient $-3/10$. Equation $y + 1 = -3/10(x - 7/2)$. At $y = 0$: $1 = -3/10(x - 7/2) \Rightarrow x - 7/2 = -10/3 \Rightarrow x = 7/2 - 10/3 = 21/6 - 20/6 = 1/6$. Equidistance check: $(p - 2)^2 + 36 = (p - 5)^2 + 16$ with $p = 1/6$ gives $49/36 + 36 = \dots \checkmark$.

Investigate further: Two routes: slope-intercept vs. $PA^2 = PB^2$. The latter never needs a gradient — set $(x - 2)^2 + 36 = (x - 5)^2 + 16$ and solve. **If A and B were swapped, does the bisector change? No — the locus is symmetric.**

3. Two circles have the same radius r . Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which line is the common chord PQ ? (adapted from TMUA 2021 Paper 1 Q1)

- A) $5x - 3y = 1$
 B) $5x - 3y = 4$
 C) $5x + 3y = 1$
 D) $3x - 5y = 4$

Answer: B) $5x - 3y = 4$

Method: Subtract: $(x+2)^2 + (y-1)^2 - r^2 = 0$ and $(x-3)^2 + (y+2)^2 - r^2 = 0$. x^2 and y^2 cancel: $(4x+4-2y+1) - (-6x+9+4y+4) = 10x-6y-8=0$, so $5x-3y=4$. The r^2 terms cancel too — the radical axis needs no radius.

Investigate further: This is the radical-axis trick: subtraction kills $x^2 + y^2$ in one line. **If the radii differed, would the same subtraction still give a line? Yes, but not the chord — it would be the radical axis, still straight. Where does the chord meet the line of centres? At its midpoint.**

4. For which values of p does $x^2 - 2px + y^2 - 6y - p^2 + 8p + 9 = 0$ represent a (real) circle? (after TMUA 2022 Paper 1 Q2)
- A) $p < -1$ or $p > 9$
 B) $-1 < p < 9$
 C) $0 < p < 4$
 D) $p < 0$ or $p > 4$

Answer: D) $p < 0$ or $p > 4$

Method: Complete squares: $(x-p)^2 - p^2 + (y-3)^2 - 9 - p^2 + 8p + 9 = 0 \Rightarrow (x-p)^2 + (y-3)^2 = 2p^2 - 8p = 2p(p-4)$. Radius² > 0 needs $p(p-4) > 0$, so $p < 0$ or $p > 4$. At $p = 0, 4$ radius 0 (a point).

Investigate further: The y -part $(y-3)^2$ contributes 9 then -9 ; the x -part contributes p^2 . Keep the p^2 terms separate — $2p^2 - 8p$ factorises cleanly. **What is the locus of centres $(p, 3)$ as p varies? The horizontal line $y = 3$.**

5. The points $A(1, 4)$, $B(4, 7)$ and $C(10, k)$ are collinear. The line through C perpendicular to AB meets the y -axis at D . What is the y -coordinate of D ?
- A) 23
 B) 24
 C) 25
 D) 26

Answer: A) 23

Method: AB gradient = 1, so $k = 13$ (since $(10-1) \cdot 1 + 4 = 13$). Perp gradient -1 through $C(10, 13)$: $y - 13 = -(x - 10)$. At $x = 0$, $y = 23$.

Investigate further: The collinearity $k = 13$ is the hidden first step; the second step is the intercept. **If AB had gradient 2, the perp gradient would be $-\frac{1}{2}$ and D would be 18 — the 1 here is the special case where the numbers stay integral.**

6. The curve $y = 4x^2$ is translated by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$, then reflected in the x -axis, then stretched parallel to the x -axis with scale factor 2. What is the equation of the final curve? (*adapted from TMUA 2020 Paper 1 Q10*)

- A) $y = -x^2 + 12x - 31$
 B) $y = -x^2 + 12x - 41$
 C) $y = -16x^2 + 48x - 31$
 D) $y = 16x^2 - 48x + 41$

Answer: A) $y = -x^2 + 12x - 31$

Method: Start $y = 4x^2$. Translate: $(x, y) \rightarrow (x - 3, y + 5)$ gives $y + 5 = 4(x - 3)^2$. Reflect $y \rightarrow -y$: $-y + 5 = 4(x - 3)^2$ or $y = -4(x - 3)^2 + 5$. Stretch $x \rightarrow x/2$: $y = -4(x/2 - 3)^2 + 5 = -4(x^2/4 - 3x + 9) + 5 = -x^2 + 12x - 31$.

Investigate further: Do the three maps on the equation, not on points. The x -stretch $(x, y) \rightarrow (2x, y)$ sends $y = f(x)$ to $y = f(x/2)$. **If the order were stretch then reflect, would the result change? Yes — transforms do not commute.**

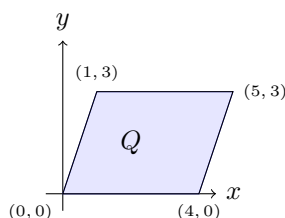
7. The circles $(x + 4)^2 + (y + 1)^2 = 64$ ($r_1 = 8$) and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (*adapted from TMUA 2019 Paper 1 Q6*)

- A) 8
 B) 16
 C) 26
 D) 42

Answer: B) 16

Method: Centre distance $d = \sqrt{(12)^2 + 5^2} = 13$. One point $\Rightarrow$ internally or externally tangent: $d = |r \pm r_1|$, so $r = |13 \pm 8|$ gives $r = 21$ or $r = 5$, difference 16.

Investigate further: The two tangencies are on opposite sides of C_1 . **If d had been 8, the two r would be 0 and 16 — $r = 0$ is a degenerate point-circle, not allowed ($r > 0$).**



8. The quadrilateral Q has vertices $(0,0)$, $(4,0)$, $(5,3)$ and $(1,3)$ in that order (a parallelogram). A second quadrilateral Q' is obtained by reflecting Q in the line $y = x$. What is the area of the overlap $Q \cap Q'$?

- A) 6
 B) 8
 C) 9
 D) 12

Answer: A) 6

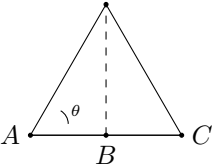
Method: Q : $0 \leq y \leq 3$, $y \leq x \leq y + 4$ (from $x = y$ to $x = y + 4$). Q' swaps x, y : $0 \leq x \leq 3$, $x \leq y \leq x + 4$. Intersection satisfies both, polygon with vertices $(0, 0), (3, 1), (3, 3), (1, 3)$. Shoelace gives area 6. Equivalently, Q area 12, overlap is half by symmetry of $y = x$.

Investigate further: Reflection in $y = x$ swaps the inequalities. **If Q were the unit square $[0, 1] \times [0, 1]$, its reflection is itself and overlap is 1 — here the 1-unit shear makes the overlap a pentagon of half the original area.**

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.

1.  In $\triangle ABC$, $AB = 10$, $BC = 7$ and $\angle A = \theta$. There are two distinct non-congruent triangles T_1 (larger area) and T_2 (smaller area) satisfying these data. It is known that $\text{Area}(T_1) = 3 \text{Area}(T_2)$. Find $\cos \theta$. (adapted from TMUA 2018 Paper 1 Q19 / BMO1 2015 Q3)
- A) $\frac{5}{7}$
 B) $\frac{\sqrt{17}}{5}$
 C) $\frac{\sqrt{51}}{8}$
 D) $\frac{\sqrt{34}}{8}$

Answer: B) $\frac{\sqrt{17}}{5}$

Method: Put $A(0, 0)$, $B(10, 0)$, C with $AC = b$, $BC = 7$, $\cos \theta$ unknown: $49 = 100 + b^2 - 20b \cos \theta$, so $b^2 - 20b \cos \theta + 51 = 0$. Two b 's are b_1, b_2 with $b_1 b_2 = 51$, $b_1 + b_2 = 20 \cos \theta$. Areas = $5b \sin \theta$, ratio $b_1/b_2 = 3$, so $b_2^2 = 17$, $b_2 = \sqrt{17}$, $b_1 = 3\sqrt{17}$, sum $4\sqrt{17} = 20 \cos \theta$, hence $\cos \theta = \sqrt{17}/5$.

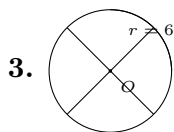
Investigate further: The quadratic in b is the ambiguous case (SSA). **If the area ratio were 2 not 3, then $2b_2^2 = 51$ would give $\cos \theta = \frac{3\sqrt{34}}{20}$ — the same structure, different surd.**

2. The square $MNOP$ has perimeter 40 (so side 10). Points R, S, T, U lie on MN, NO, OP, PM respectively with $MR = x$ and $RSTU$ a rectangle whose sides are at 45° to the square's sides. For $0 < x < 10$, the area of $RSTU$ is $A(x) = -2x^2 + 20x$ (up to translation). What is the largest x for which $A(x) = 20$? (after TMUA 2023 Paper 1 Q5)
- A) $5 + \sqrt{5}$
 B) $5 + \sqrt{15}$
 C) $5 + \sqrt{20}$
 D) $10 - \sqrt{5}$

Answer: B) $5 + \sqrt{15}$

Method: $-2x^2 + 20x = 20 \Rightarrow x^2 - 10x + 10 = 0 \Rightarrow x = 5 \pm \sqrt{15}$. Larger is $5 + \sqrt{15}$. The quadratic $A(x)$ is symmetric about $x = 5$ (midpoint of side).

Investigate further: The area as a quadratic in x comes from $A = 2x(10 - x)$. **The maximal area is at $x = 5$, $A = 50$; $A = 20$ cuts the parabola twice, symmetric about the vertex.**



3. A circle has centre O and radius 6. Points P, Q, R lie on the circle with $\angle POQ \geq \frac{\pi}{2}$ and $[POQ] = 9\sqrt{3}$ (area). What is the *greatest possible* area of $\triangle PRQ$? (adapted from TMUA 2022 Paper 1 Q14)

- A) $18 + 9\sqrt{3}$
 B) $27\sqrt{3}$
 C) $27 + 9\sqrt{3}$
 D) $36 + 9\sqrt{3}$

Answer: B) $27\sqrt{3}$

Method: $[POQ] = \frac{1}{2}r^2 \sin \angle POQ = 18 \sin \theta = 9\sqrt{3} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}$, so $\theta = 60^\circ$ or 120° , but $\geq 90^\circ$ forces 120° . For fixed chord PQ , $[PRQ]$ is maximal when R is opposite the midpoint of arc PQ , height $= r + r \cos(\theta/2) = 6 + 3 = 9$, base $PQ = 6\sqrt{3}$, so $\max = \frac{1}{2} \cdot 6\sqrt{3} \cdot 9 = 27\sqrt{3}$.

Investigate further: The 120° chord has length $r\sqrt{3}$. **If θ had been 60° , the greatest $[PRQ]$ would be $9 + 18 = 27$ with different geometry — the $\geq \pi/2$ filters the case.**

4. Let $P(p, q)$ and circle $C : x^2 + 2fx + y^2 + 2gy + h = 0$ with centre $C_0 = (-f, -g)$ and radius $r = \sqrt{f^2 + g^2 - h}$. Let $L = |PC_0|$. Which data set is *minimal sufficient* to determine L ? (adapted from MAT 2020 Q1H / TMUA 2022 Paper 2 Q4)

- A) f, g, h
 B) f, g, p, q
 C) f, h, p, q
 D) g, h, p, q

Answer: B) f, g, p, q

Method: $L^2 = (p + f)^2 + (q + g)^2$ uses p, q and f, g only; h (hence r) is irrelevant to centre-point distance. Any set missing f or g or p or q is insufficient, and $\{f, g, p, q\}$ already suffices, so it is minimal.

Investigate further: Many confuse L with PT (tangent length $\sqrt{L^2 - r^2}$) which *does* need h . **If the question had asked for the length of the tangent from P to C , the answer would need h as well.**

5. Circle $C_1 : x^2 + y^2 = 25$ is fixed. A second circle C_2 has radius 4 and centre (a, b) uniformly random in the rectangle $-2 \leq a \leq 2$, $-3 \leq b \leq 3$ (so the centre is equally likely to be anywhere in that 4×6 rectangle). What is the probability that C_1 and C_2 intersect (in two points, touching counts as intersecting)? (after TMUA 2022 Paper 1 Q19)

- A) $\frac{9}{25}$
 B) $\frac{16}{25}$
 C) $\frac{16-\pi}{24}$

D) $\frac{24-\pi}{24}$ **Answer:** D) $\frac{24-\pi}{24}$

Method: Centres distance $d = \sqrt{a^2 + b^2}$. Intersect iff $|5 - 4| \leq d \leq 9$, i.e. $1 \leq d \leq 9$. Max d in rectangle is $\sqrt{13} < 9$, so upper bound never fails; need $d \geq 1$. Complement $d < 1$ is disc radius 1 area π , fully inside 4×6 rectangle, so $P = 1 - \pi/24$.

Investigate further: Touching ($d = 1$ or 9) counts, so inequality is inclusive. **If the rectangle were $[-5, 5] \times [-5, 5]$, the $d \leq 9$ would also matter and the probability would be $(\pi \cdot 81 - \pi)/100$. Here the rectangle is small enough that only the inner exclusion matters.**

Top 5 Patterns — Worksheet #1

- 1. Implicit line $\rightarrow$ gradient: rearrange to $y = mx + c$ (or read $-a/b$).** The constant c never touches m — strip it off last. (see A1, A7, B5)
- 2. Perpendicular gradients are negative reciprocals:** $m_1 m_2 = -1$. Slack jaw never flips and negates; keep both operations. (see A3, B1, B8, C5, C8)
- 3. Perpendicular bisector = midpoint + negative-reciprocal slope.** Always hit the midpoint first; the bisector is the line through it with the flipped-inverted gradient. (see B3, C2, D1)
- 4. Perpendicular distance formula:** $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$. Numerator from substitution, denominator from the normal vector's length. (see A6, B2, B6, C7, D3)
- 5. Shoelace (signed) area:** $\frac{1}{2} |\sum x_i y_{i+1} - \sum y_i x_{i+1}|$. Vertices in cyclic order; the $\frac{1}{2}$ and the absolute value are non-negotiable. (see A10, B4, B10, D5)

Common Traps — Worksheet #1

- 1. Sign of the implicit gradient.** $6x - 3y + 15 = 0$ has gradient 2, not -2 and not 3 — always solve for y explicitly, or read $-a/b$ with both signs. (see A1, A7, B1)
- 2. Distance: absolute value & square root together.** $\frac{|\dots+c|}{\sqrt{\dots}}$ — dropping the absolute value floors the answer at zero and loses the sign of the offset. (see A6, B2, C7)
- 3. Perpendicular does not mean “negated slope”.** $m = 2$ pairs with $-\frac{1}{2}$, not -2 ; only the flip-invert passes $m_1 m_2 = -1$. (see C5, C8, B1)
- 4. Shoelace: half the absolute difference, vertices in order.** Out-of-order vertices produce self-crossing regions whose signed sum misleads — keep cyclic order and take half of the magnitude. (see B4, B10, D5)
- 5. Collinearity: compare the right gradient pairs.** Three points are collinear iff *every* pair gives the same slope — one matching pair is never enough. (see A9, C6)

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths